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# TALKPLAY: Multimodal Music Recommendation with Large Language Models

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A demo is available at: `https://talkpl.ai/`

## Abstract

We present TALKPLAY, a novel multimodal music recommendation system that reformulates recommendation as a token generation problem using large language models (LLMs). By leveraging the instruction-following and natural language generation capabilities of LLMs, our system effectively recommends music from diverse user queries while generating contextually relevant responses. While pre-trained LLMs are primarily designed for text modality, TALKPLAY extends their scope through two key innovations: a *multimodal music tokenizer* that encodes audio features, lyrics, metadata, semantic tags, and playlist co-occurrence signals; and a *vocabulary expansion* mechanism that enables unified processing and generation of both linguistic and music-relevant tokens. By integrating the recommendation system directly into the LLM architecture, TALKPLAY transforms conventional systems by: (1) unifying previous two-stage conversational recommendation systems (recommendation engines and dialogue managers) into a cohesive end-to-end system, (2) effectively utilizing long conversational context for recommendation while maintaining strong performance in extended multi-turn interactions, and (3) generating natural language responses for seamless user interaction. Our qualitative and quantitative evaluation demonstrates that TALKPLAY significantly outperforms unimodal approaches based solely on text or listening history in both recommendation performance and conversational naturalness.

## 1 Introduction

The explosive growth of music streaming platforms has created overwhelming choices for users, with millions of tracks available [2]. Traditional recommendation systems [22, 28, 42, 6] rely heavily on listening history rather than explicit user queries, while text-to-music retrieval systems [48, 24, 33, 15, 14] only support single-turn interactions, creating limitations when users are dissatisfied with initial results and need to refine their requests. Conversational recommendation systems (CRS) [18, 10, 51, 9, 25] address these limitations by enabling natural language interactions for item discovery, capturing user preferences, providing relevant recommendations, and generating responses.

In parallel with these developments, recent advancements in large language models (LLMs) [1, 44, 19] have demonstrated remarkable capabilities across diverse domains, from natural language understanding to complex reasoning tasks. Their ability to follow instructions, generate contextually relevant responses, and leverage extensive pre-trained knowledge has led to increasing exploration of LLMs in recommendation systems [20, 31]. However, a key challenge is the semantic gap between LLMs and recommender systems, as items are typically indexed by unique identifiers outside the LLM's vocabulary [23], making direct application difficult. More recent approaches address this by

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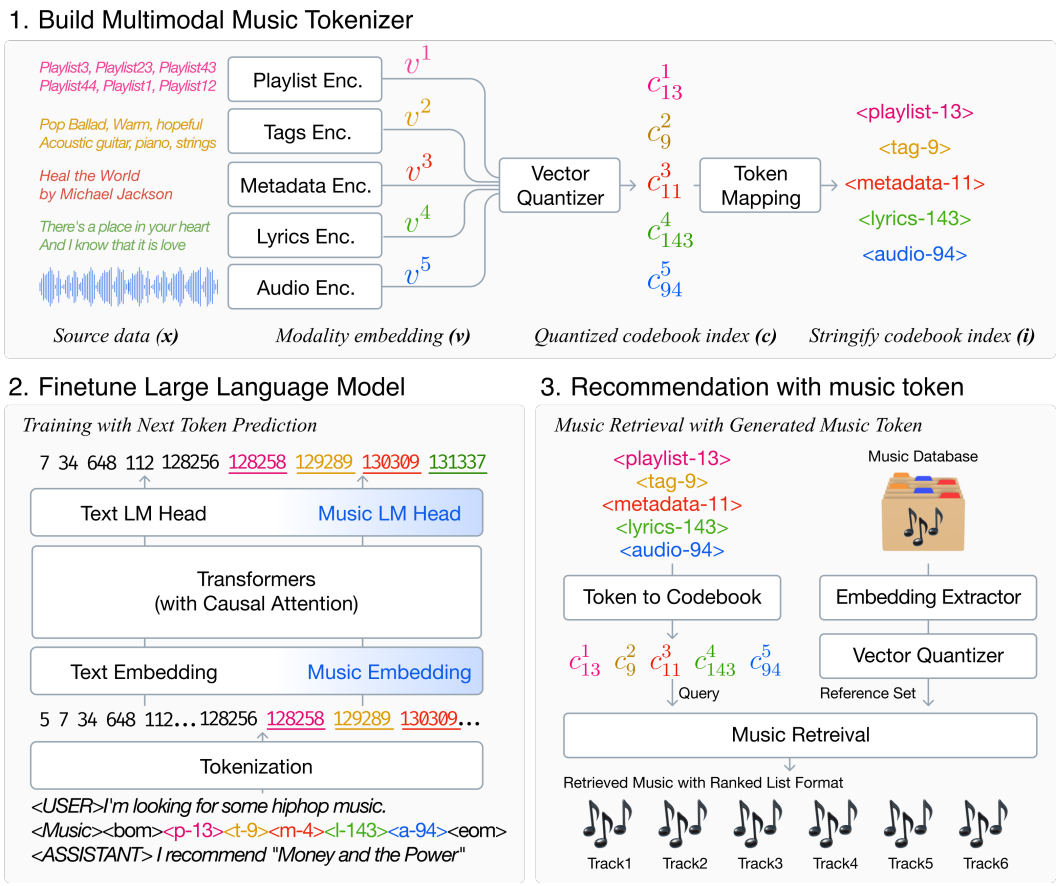


Figure 1: Overview of TALKPLAY: (1) The multimodal music tokenizer converts source data into modality embeddings $v$ and quantizes them into codebook indices $c$, which are mapped to music tokens $i$. (2) The LLM is fine-tuned on text and music token sequences. (3) The generated music tokens are mapped back to codebook indices and are used as queries to retrieve music items from the database.

fine-tuning LLMs on recommendation data [53, 50, 17], enabling them to learn associations between natural language queries and item representations. However, the seamless integration of multimodal content with LLMs remains largely unexplored.

To address this problem, we present TALKPLAY, a multimodal music recommendation system powered by LLMs. Our approach reformulates recommendation as next token prediction, also known as generative retrieval [43, 46, 39], through novel *music tokenization* and *vocabulary expansion* techniques. Our system represents music items as token sequences that encode audio features, lyrics, metadata, tags, and playlist co-occurrence, enabling direct recommendations through next-token prediction within conversations. This integrated approach offers several advantages: end-to-end learning without explicit dialogue management, inheritance of broad musical knowledge from pre-trained LLM weights, and effective utilization of multiple information sources (user preferences, queries, interaction patterns) through the LLM architecture.

Our experimental results demonstrate that: (1) the proposed multimodal recommendation system outperforms unimodal approaches based solely on text or listening history; (2) our LLM-based generative retrieval system maintains strong recommendation performance even in extended multi-turn contexts; and (3) compared to two-stage systems, our approach shows superior performance in human evaluations of recommendation relevance and response generation quality.

## 2 TALKPLAY: The Proposed System

As illustrated in Figure 1, TALKPLAY consists of two parts: a music tokenizer and a large language model. The music tokenizer encodes multimodal music embeddings into tokens. A single music item is represented by multiple tokens, each of which represents a different modality. The model is a multimodal LLM that is presumably: i) capable of modeling music recommendation conversations (with text and music tokens), and ii) pretrained with a large amount of data including musical corpus.

### 2.1 Music Tokenizer

The music tokenizer in TALKPLAY is designed under following goals and constraints. First, to retrieve and recommend items, the model needs to be connected to the database. Second, to easily utilize LLM as a recommendation engine, the items in the database need to be discretized as tokens. Third, the embeddings of those item tokens need to be trained well, requiring some minimum token occurrence in the training data. Given these conditions, we propose the following scheme for music tokenization in TALKPLAY. A music item $x_i$ ($i$ is an item index) may have rich information including (1) playlist co-occurrence, (2) semantic tag information, (3) metadata, (4) lyrics, and (5) audio.

**Item Embeddings**: For all the data modalities, pretrained models are used to extract item embedding $v_i^m$, where $m$ stands for a modality index (1-5 for playlist co-occurrence, semantic tag, metadata, lyrics, and audio, respectively.) For playlist co-occurrence, a word2vec-style [2] item embedding model is trained using the training dataset ($v_i^1 \in \mathbb{R}^{128}$). For semantic information, metadata, and lyrics, NV-Embed-v2 [29] is chosen as its state-of-the-art performance in various text embedding benchmarks [34] ($v_i^2, v_i^3, v_i^4 \in \mathbb{R}^{4096}$). For audio, the MusicFM [47] model is chosen for its strong performance in many music informatics tasks ($v_i^5 \in \mathbb{R}^{1024}$).

**Quantization**: Due to the long-tail nature of music consumption and the limited size of available training data, all the item embeddings $v_i^m$ are vector-quantized into $K$ clusters (per modality) using K-means clustering. Here, $c_k^m$ represents a cluster centroid. In this paper, we chose $K = 1024$. As a result, the overall music database is represented by $K \times M$ indices, where the total number of modalities $M$ is 5. However, this does not mean that all music items are reduced to $1024 \times 5 = 5120$ unique clusters. A music item is represented by 5 modality tokens, where each modality embedding has 1024 clusters. Theoretically, the proposed music tokenization scheme is capable of representing $1024^5$ = about 1,126 trillion unique music items.

**Token Mapping**: In this final step of tokenization step, the multimodal cluster indices of each music item is used in our token format: `<|modality-index|>`. For example, a music item whose embeddings are mapped to 29th audio cluster, 2nd metadata cluster, 98th lyrics cluster, 3th semantic cluster, and 59th user-item cluster is converted to `<|playlist-59|><|semantic-361|><|metadata-7|><|lyrics-98|><|audio-29|>`.

### 2.2 LLM: The Recommender

The LLM, our recommendation engine, is based on a pretrained model. We opt for instruction finetuned model to utilize the conversationability as our goal is to build a conversational recommender. Our task is highly specific compared to the scope of usual LLM training, implying relatively small models might be enough. Altogether, Llama-3.2-1B-Instruct is chosen for our base model [16].

To enhance the LLM's understanding of multimodal music information, we expand its vocabulary to include multimodal music tokens [52]. In detail, the vocabulary size of the base model is expanded by $1024 \times 5 = 5120$, to include the multimodal music tokens. Two more special tokens, `<start_of_music>` and `<end_of_music>`, are added as well, since they are used to wrap music tokens in the training data. This modification means 5122 new embeddings ($e_k^m$ and two special token embeddings) to the model, each of which is 2048-dim in the Llama-3.2-1B model. Equivalently, it adds about 10.5M parameters to the model, increasing the model size by less than 1%. In the synthetic conversation, music items are represented with the multimodal music tokens, which are tokenized using the added vocabulary and fed to the model. This allows the model to directly understand and generate multimodal music tokens along with regular text tokens during conversations.

### 2.3 Training

TALKPLAY employs a principled supervised finetuning approach for LLMs, leveraging autoregressive next token prediction with cross-entropy loss. We transform synthetic conversational data (Section 3.2) by systematically replacing track identifiers with their corresponding multimodal music tokens. Formally, given a token sequence $x = \{(q_1, m_1, r_1), ..., (q_t, m_t, r_t)\}$ representing a conversation with user queries $q$, recommended music tokens $m$, and system responses $r$, the model is trained to predict each subsequent token in the sequence. Each token $t$ can be either a text token from user queries or system responses, or a music token from recommended items. We use the standard autoregressive cross-entropy loss: $\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} \log P(t_i|t_1, t_2, ..., t_{i-1})$, where $P(t_i|t_1, t_2, ..., t_{i-1})$ represents the model's predicted probability of token $t_i$ given its preceding context.

This unified formulation enables the model to first predict contextually appropriate music tokens based on the user query $p(m|q)$, and subsequently generate coherent responses conditioned on both the query and predicted music tokens $p(r|q, m)$. Building on insights from prior research [11], we leverage the inherent coarse-to-fine structure of music modalities by predicting tokens in a carefully chosen sequence: *playlist - semantic - metadata - lyrics - audio* tokens. We validate this ordering strategy through comprehensive ablation studies on feature importance (Section 5.3).

The central goal of this approach is to effectively learn the newly introduced multimodal music token embeddings $e_k^m$ by capturing rich contextual relationships between conversations and music items. We initialize these music token embeddings using the statistics of the base model's existing embeddings, following the vocabulary expansion method proposed by Hewitt et al. [21], $e_k^m \sim \mathcal{N}(\mu_E, \Sigma_E)$, where $\mu_E$ and $\Sigma_E$ represent the mean and covariance of the base model's token embeddings.

### 2.4 Inference: Recommendation by Generation

During inference, we use typical LLM generation parameters (temperature=1.0, top-p=0.9, repetition penalty=1.0). The music tokens are generated before the assistant's response message to ensure recommendations are based solely on the query, avoiding potential popularity biases that could arise from conditioning on generated text [20].

The generated tokens must then be mapped back to music items through a reverse lookup of the tokenization process described in Section 2.1. While each individual cluster codebook index has a many-to-one track-to-cluster relationship, our tokenizer creates sequences of five multimodal tokens that form a nearly one-to-one track-to-cluster-sequence mapping (similar to Semantic IDs [39] using residual vector quantization), making the process more deterministic than typical approximate nearest neighbor searches used in industry recommenders [3].

Given a sequence of five multimodal tokens representing a recommendation, we select items matching the token indices across modalities. However, due to the vast token space (1,126 trillion combinations) relative to our dataset size, exact matches are often unavailable. In such cases, we fall back to partial matching using token overlap scoring with feature importance weights assigned to each modality. The final score is determined as $s_{\text{retrieval}} = \sum_{i \in \{\text{modalities}\}} \lambda_i s_i$, where each $s_i$ term is a binary indicator (0 or 1) representing whether there is an overlap in the corresponding modality. Based on our preliminary experiment, we use a quadratically decreasing weighting scheme that best captures the relative importance of different music features. The final weights used are $\lambda_{\text{cf}} = 25$, $\lambda_{\text{semantic}} = 16$, $\lambda_{\text{metadata}} = 9$, $\lambda_{\text{lyrics}} = 4$, and $\lambda_{\text{audio}} = 1$. This weighting scheme reflects our finding that playlist co-occurrence token provide the strongest recommendation performance gain, followed by semantic tags, metadata, lyrics, and audio features. The results of these weighting experiments are presented in detail in Section 5.3.

## 3 Data Generation

It is prohibitively difficult to perform research on recommendation without access to real-world data – information about items, users, and their interactions. Another challenge lies in the fact that the conversational recommendation system is in its early stage and so the publicly available dataset is scarce. The Conversational Playlist Curation Dataset (CPCD) is probably the most relevant one, but it comes with 917 conversations which are not sufficient to train a large model [7].